Paleomagnetism of the Kau Volcanic Series, Hawaii

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Paleomagnetic studies of 54 superposed lavas from the northwest wall of Mokuaweoweo, Mauna Loa, Hawaii, show that changes in direction of the geomagnetic field were very small during the time of extrusion of these lavas. After reasonable estimates are made for the time involved (a period of several thousands of years, occurring sometime after 10,000 years ago), it is tentatively concluded that secular variation due to nondipole field changes in the central Pacific Ocean is very small and can be described by an angular deviation for virtual poles of only 0.05 radians (3°). In combination with other Hawaiian paleomagnetic studies representing longer time intervals, this leads to an estimated angular deviation of 0.20 radians (12°) for 'wobble' of the main dipole during the Brunhes magnetic polarity epoch.

INTRODUCTION

Studies of the behavior of the geomagnetic field in the central Pacific Ocean are of interest because this area is presently unique, being the only region where the nondipole portion of the geomagnetic field is essentially missing [*Cox*, 1962]. Moreover, as noted several decades ago by *Fisk* [1931], the secular variation (rate of change of the geomagnetic elements) is also anomalously low in this region. Studies by paleomagnetic means of the duration of this 'anomalous' feature have shown that it has persisted since at least about 1750 [*Doell and Cox*, 1963], and also possibly for most of the last half million years or so [*Doell and Cox*, 1965*a*]. The present work, a paleomagnetic study of the Kau volcanic series of the active volcano Mauno Loa, is an extension of the last-mentioned study. These prehistoric lavas, like those previously studied, cannot be assigned accurate ages, nor is the time interval between successive flows known. Thus, the way in which the directions of their natural remanent magnetization represent those of the ancient field is known only as a sequence. However, certain estimates of time can be made. As with the previous studies of Hawaiian lavas, the paleomagnetism of the Kau volcanic series lavas suggests a very subdued nondipole field in the central Pacific during the last several thousands of years.

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Samples of the Kau volcanic series lavas used in this study were taken from its type locality on the northwest wall of Mokuaweoweo, the summit caldera on the volcano Mauna Loa. A total of 54 superposed lavas very near the location of the section measured by *Stearns and MacDonald* [1946, p. 190], latitude 19.486°N, longitude 204.400°E, were sampled. At this locality, (Figure 1) the caldera wall is about 140 meters high and is accessible throughout this height by working up a large slump block just northeast of the section sampled. This location is marked on Figure 1 in reference to other features of the summit region; further physiographic details may be found on the Mauna Loa, Hawaii quadrangle [U. S. Geological Survey, 1966].

A brief physical description of the lavas sampled is given in Figure 2 by means of a schematic section of the volcanic materials encountered. With rare exceptions, all of the lavas sampled are separated from one another by layers of volcanic rubble, scoria blocks, or other evidence of a time separation between emplacement. Very few of the lavas sampled are laterally continuous for large distances, most of them terminating within several tens of meters to a few hundreds of meters of the sampled section. Only one dike was observed nearby in the caldera walls; it leads to unit number 13 from below the caldera floor. This unit may in fact be a sill rather than a subaerially erupted

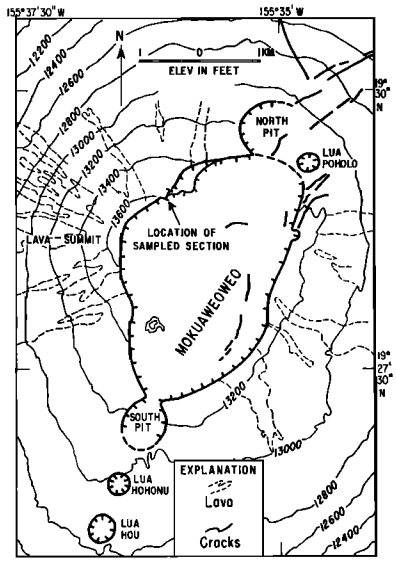


Fig. 1. Sampling site location with respect to other features on the summit of Mauna Loa.

lava flow, as may other lavas in the section. However, as will be discussed below, it is very unlikely that more than a few of the dense lava layers sampled are intrusions. Further details on the geology of the Kau volcanic series may be found in *Stearns and MacDonald* [1946, pp. 76–96] or in *MacDonald* [1969].

Every accessible coherent layer of lava more than a few meters in lateral extent was sampled by means of a portable water-cooled diamond coring drill [*Doell and Cox*, 1967*a*]. Eight 2.5-cm diameter by 10- to 20-cm-long cores were drilled from each lava and oriented in geographic coordinates before separation from the rock exposures. The average thickness of the different lavas sampled can be scaled from Figure 2, and the lateral distance between the farthest removed cores of a given lava is listed after the lava number to the right of the schematic section.

Laboratory Procedures

Each of the 400-odd samples collected from the Kau volcanic series were cut into specimens 2.25 cm long for measurement of magnetic properties. These properties were natural remanent magnetization (NRM), measured on a spinner-type magnetometer [*Doell and Cox*, 1965*b*], saturation magnetization versus temperature [*Doell and Cox*, 1967*b*], and stability in alternating demagnetizing fields [*Doell and Cox*, 1967*c*]. The NRM of one specimen from each core was measured, and one specimen from each lava flow was treated in alternating demagnetizing fields of 12.5, 25, 50, 100, 200, 400, 600, and 800 oe peak value, the remaining remanent magnetization being measured after each treatment. Saturation magnetization versus temperature measurements were made on one specimen from each lava.

A summary of the alternating field demagnetization studies on the 54 specimens involved is given in Figure 3 as a set of histograms of the intensity of remanent magnetization (in logarithmic intervals) remaining after each treatment. The values of remanent magnetization remaining after treatments at 50, 100, 200, 400, and 800 oe peak value are plotted so that the 'zero' positions of the histograms represent the logarithms of the peak alternating field values along the 'abscissa' of the illustration. The NRM histogram of these same sample specimens is shown for comparison. One further comment is necessary: each of the individual specimen demagnetization curves showed a monotonically decreasing intensity with increasing demagnetizing field. In fact, the shape of the average shown by the dashed line in Figure 3 is typical of most samples. As may be seen, the Kau volcanic series lavas have relatively high stability against alternating field demagnetization, similar to that of the Hawaiian lavas previously studied [*Doell and Cox*, 1963, 1965*a*]. Moreover, throughout the demagnetization experiments, the direction of remanent magnetization within a given specimen did not deviate from its NRM direction beyond the precision of the measuring equipment.

Because of the high magnetic stability of the NRM in these lavas, magnetic 'cleaning' experiments were not carried out, and the NRM directions were used to define the directions of the geomagnetic field at the times the lavas cooled.

In passing, it is interesting to note that the histograms in Figure 3 become more sharply peaked at the higher alternating field values. This seems to suggest that the variation in NRM intensity from lava to lava is mostly

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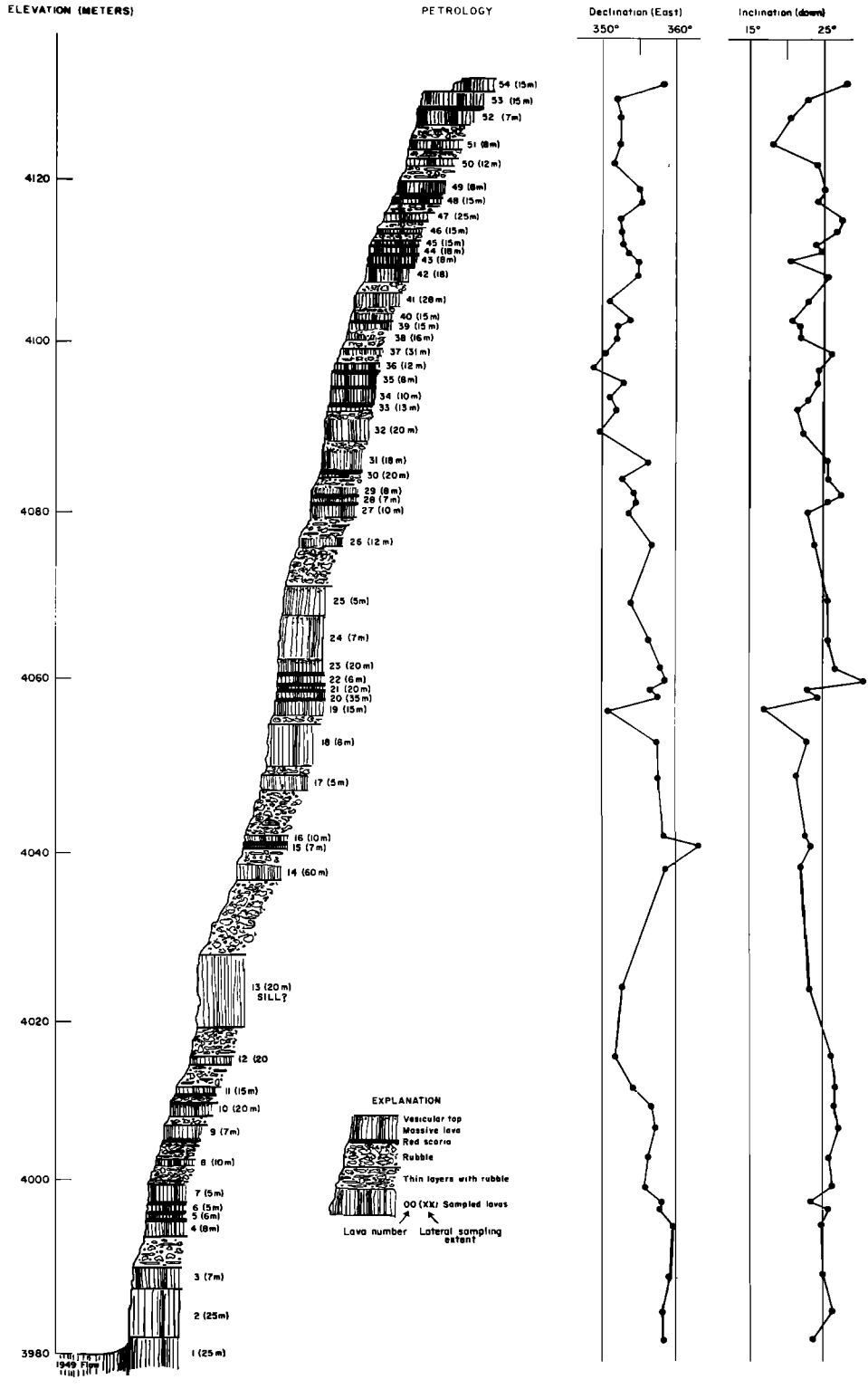


Fig. 2. Schematic section of lava flows exposed in the northwest wall of Mokuaweoweo with paleomagnetic declination and inclination measured in these lavas.

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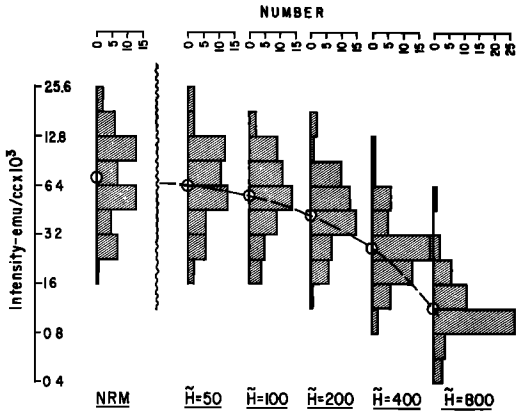


Fig. 3. Histograms of remanent magnetization in lavas of the Kau volcanic series after partial alternating field demagnetization at peak field values $\tilde{H}$. Dashed curve connects mean values.

due to variations in the lower portion of the coercive force spectrum.

The saturation magnetization versus temperature measurements were made in constant fields between 1500 and 4000 oe, whatever level was necessary to approach saturation in a given sample. The heating was carried out in air. Curie temperatures (Table 1) ranged between 500° and 575°C, with the exception of the topmost lava (200°C). Marked chemical change during heating was noted in only four lavas. The high Curie temperatures generally preclude remagnetization by superjacent lavas, and there is no evidence that the section sampled has been heated sufficiently for general remagnetization to have occurred.

The problem of paleomagnetic accuracy (i.e., the problem of determining how closely the direction of remanent magnetization in a lava flow corresponds to the geomagnetic field direction at the time the lava cooled) has previously been investigated in detail for lavas erupted on Hawaii [*Doell and Cox,* 1963, 1965*a*]. In all physical respects the lavas of the Kau volcanic series are similar to those previously studied, and their alternating field demagnetization spectra as well as saturation magnetization versus temperature characteristics are also the same. Thus, the conclusions previously drawn [*Doell and Cox,* 1965*a*] should also apply to the Kau volcanic series. These conclusions were, '(1) The NRM measured in the Hawaiian lavas is predominately TRM acquired when the lavas cooled. (2) This NRM was acquired essentially parallel to the directions of the geomagetic field at the time the lavas cooled. (3) The NRM of these lavas has not changed significantly since it was acquired.' It was also shown [*Doell and Cox,* 1963] that the largest source of scatter within a lava occurred at the core-to-core level and, hence, little increase in accuracy can be obtained by measuring more than one specimen from each core.

The results of the NRM measurements are given in Table 1 as averages for each lava flow; these are the average direction and intensity plus certain statistical parameters and the positions of the corresponding virtual geomagnetic poles (VGP). These flow-average directions, therefore, may be taken as the best estimates

TABLE 1. Flow-Average NRM Data

Lava No.	N	Decl. of J_N, deg	Incl. of J_N, deg	$J_N \times 10^3$ ± s.d.	R	k	α_{95}, deg	S, deg	Lat. of VGP, deg	Long. of VGP, deg	θ_c, °C
54	8	358.2	28.1	9.8 ± 4.9	7.9603	176	4.2	6.1	85.1	45.3	200
53	8	351.9	22.8	12.2 ± 4.3	7.9789	332	3.0	4.4	79.1	71.2	(350)
52	8	352.3	20.4	12.6 ± 4.5	7.9927	956	1.8	2.6	78.4	65.1	525
51	8	352.3	18.0	11.0 ± 2.7	7.9623	185	4.1	6.0	77.3	61.6	500
50	7	351.5	24.0	11.9 ± 4.3	6.9876	482	2.8	3.7	79.3	75.3	575
49	8	355.0	25.0	12.1 ± 6.0	7.9733	262	3.4	5.0	82.1	62.0	575
48	7	355.2	24.1	10.3 ± 2.6	6.9895	571	2.5	3.4	81.7	58.6	500
47	8	352.2	27.6	10.5 ± 3.8	7.9913	808	2.0	2.8	81.1	82.6	575
46	8	352.4	26.9	15.3 ± 3.4	7.9951	1436	1.5	2.1	81.1	79.6	525
45	7	352.9	23.9	10.1 ± 2.2	6.9798	297	3.5	4.7	80.2	69.7	525

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TABLE 1. (*continued*)

Lava No.	N	Decl. of J_N, deg	Incl. of J_N, deg	$J_N \times 10^3$ ± s.d.	R	k	α_{95}, deg	S, deg	Lat. of VGP, deg	Long. of VGP, deg	θ_c, °C
44	8	353.5	24.8	7.8 ± 3.0	7.9836	428	2.7	3.9	81.0	69.3	575
43	7	354.9	20.3	11.1 ± 4.3	6.9516	124	5.4	7.3	79.7	53.8	500
42	8	354.8	25.7	5.2 ± 2.0	7.9887	620	2.2	3.3	82.2	65.1	550
41	8	350.8	22.8	5.1 ± 2.2	7.9820	389	2.8	4.1	78.3	75.0	525
40	8	353.9	20.6	6.1 ± 2.6	7.9735	264	3.4	5.0	79.4	58.9	525
39	8	352.0	21.9	11.5 ± 6.6	7.9313	102	5.5	8.0	78.8	69.1	575
38	8	352.0	21.6	9.0 ± 2.8	7.9522	147	4.6	6.7	78.7	68.8	550
37	8	351.2	26.0	10.7 ± 4.1	7.9712	243	3.6	5.2	79.8	81.0	525
36	8	348.7	24.1	6.1 ± 1.8	7.9812	372	2.9	4.2	77.2	83.8	550
35	8	352.8	24.1	7.2 ± 1.2	7.9888	624	2.2	3.2	80.3	70.5	575
34	8	351.0	22.9	6.3 ± 2.0	7.9676	216	3.8	5.5	78.5	74.7	575
33	8	352.0	21.2	13.4 ± 1.5	7.9571	163	4.4	6.4	78.5	67.6	550
32	7	350.0	22.0	10.5 ± 4.4	6.9833	359	3.2	4.3	77.2	77.1	575
31	8	356.2	25.4	8.7 ± 3.4	7.9892	646	2.2	3.2	82.8	55.8	525
30	8	352.5	25.3	9.1 ± 6.7	7.9882	592	2.3	3.3	80.6	74.7	550
29	8	354.2	27.2	7.4 ± 2.5	7.9947	1324	1.5	2.2	82.5	72.7	575
28	8	354.8	25.5	12.9 ± 3.2	7.9680	219	3.8	5.5	82.1	64.7	500
27	7	353.4	22.9	7.3 ± 3.3	6.9804	306	3.5	4.6	80.1	65.3	575
26	8	356.6	23.6	9.6 ± 3.7	7.9700	233	3.6	5.3	82.1	49.5	575
25	8	353.6	25.3	3.4 ± 2.2	7.9883	601	2.3	3.3	81.3	70.2	550
24	8	356.0	25.3	2.7 ± 1.3	7.9729	258	3.4	5.0	82.7	56.9	575
23	8	357.7	26.3	4.4 ± 1.9	7.9828	407	2.8	4.0	84.0	45.9	575
22	7	358.6	30.2	3.2 ± 0.6	6.9937	946	2.0	2.6	86.5	47.2	550
21	8	356.1	22.7	3.7 ± 1.7	7.9561	159	4.4	6.4	81.5	50.9	575
20	8	357.6	24.5	3.3 ± 1.4	7.9557	158	4.4	6.4	83.0	43.8	550
19	8	350.6	16.7	7.2 ± 4.2	7.9691	227	3.7	5.4	75.8	65.5	550
18	8	357.5	22.8	2.1 ± 1.6	7.9854	480	2.5	3.7	82.0	42.5	550
17	8	357.5	21.1	2.4 ± 0.8	7.9615	182	4.1	6.0	81.1	40.7	575
16	5	358.1	22.4	2.5 ± 1.5	4.9842	253	4.8	5.1	81.9	37.9	575
15	7	3.1	23.2	3.5 ± 2.0	6.9848	394	3.0	4.1	82.0	1.9	575
14	8	358.5	21.7	5.6 ± 1.7	7.9856	488	2.5	3.7	81.6	34.4	575
13	7	352.7	23.1	3.5 ± 2.1	6.9968	1855	1.4	1.9	79.7	68.9	575
12	8	351.7	26.0	8.4 ± 4.2	7.9816	381	2.8	4.2	80.2	79.5	(250)
11	8	354.0	26.5	8.0 ± 3.1	7.9849	465	2.6	3.8	82.1	71.4	550
10	8	356.7	26.2	4.2 ± 1.0	7.9609	179	4.2	6.1	83.5	54.4	(300)
9	8	357.1	27.0	8.2 ± 1.9	7.9914	812	2.0	2.8	84.1	52.9	575
8	8	356.1	25.7	7.3 ± 1.4	7.9829	409	2.7	4.0	83.0	57.1	575
7	8	355.6	26.1	6.0 ± 1.3	7.9836	427	2.7	3.9	82.9	61.1	550
6	8	358.1	23.0	8.2 ± 2.5	7.9845	451	2.6	3.8	82.3	38.6	525
5	8	357.9	25.4	9.8 ± 4.2	7.9677	217	3.8	5.5	83.5	42.6	575
4	8	359.6	24.7	8.0 ± 3.5	7.9692	227	3.7	5.4	83.4	27.5	(475)
3	8	359.0	24.7	8.5 ± 2.0	7.9915	827	1.9	2.8	83.4	32.5	575
2	8	358.1	26.0	2.7 ± 0.7	7.9901	707	2.1	3.0	84.0	42.1	575
1	7	358.0	23.3	2.7 ± 0.7	6.9804	306	3.4	4.6	82.4	39.1	575

Notes.

N, number of sample cores per lava.
J_N, NRM in emu/cc.
R, vector sum of N unit vectors.
k, intraflow precision parameter = $(N - 1)/(N - R)$ [*Fisher*, 1953].
α_{95}, 95%- confidence limit [*Fisher*, 1953].
S, intraflow angular standard deviation factor = $(2N - 2\sum_{i=1}^{N} \cos \delta_i)^{1/2}/(N - 1)^{1/2}$, where δ_i is the angle between the individual core directions and the mean lava direction.
θ_c, Curie temperature. Values in parentheses are estimates of temperature at which chemical change takes place.

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of the ancient geomagnetic field directions, and the statistical parameters (for example, the 95%-confidence cone defined by α_{95} [*Fisher*, 1953]) indicate how accurately these field directions have been determined. In calculating these average directions, several instances arose in which one specimen had a direction well removed from the restricted distribution of directions of the other seven specimens (in one case, unit number 39, three specimens were widely scattered). In these cases, the aberrant data were not included in the analyses, although their inclusion would not significantly change the overall results. The 'average' accuracy of this data set is summarized in Figure 4, which is a histogram of the 95%-confidence-cone semiangles of the individual lavas.

Previously [*Doell and Cox*, 1963], it was noted that the local magnetic field direction determined by the NRM in a lava is not strictly parallel to the ancient regional field direction because of local anomalies whose extent is larger than that sampled by the flow. For the present discussion, it is of more interest to compare one direction to another, and we assume that this anomaly would be essentially constant during the extrusion of this section of lavas; thus, the correction for individual flows has not been made. If the correction had been made for these data, the minimum 95%-confidence-cone semiangle about the ancient *regional* field direction, γ_{95}, would be estimated as 2.5° and the maximum as 6.0°.

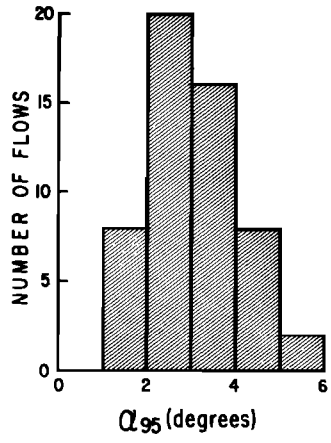


Fig. 4. Histogram of NRM 95%-confidence-cone angles of individual lava flows within the Kau volcanic series.

Analysis of Flow-Average Data

In similarity with the previously reported Hawaiian paleomagnetic data [*Doell and Cox*, 1965*a*], strict application of Fisher's statistical methods [*Fisher*, 1953] cannot be made on these data because they do not satisfy the assumption of being sequentially unrelated. That is, as is evident from the plots of declination and inclination of Figure 2, the lavas have sampled the field direction changes at too rapid a rate to satisfy the requirement that the average time interval between successive data points is longer than all the periods involved in geomagnetic field changes.

Recently, *Watson and Beran* [1967] have provided a statistical method for testing a sequence of unit vectors for serial correlation. Briefly, the test is to sum the cosines of the angles between the successive directions observed and compare this with similar sums computed from many randomly generated sequences of the same data set. The results of the test are summarized in Figure 5, and we note that the sum of the cosines with the lavas in their observed sequence exceeds, by four standard deviations, the expected sum of cosines were the data serially unrelated.

In spite of the failure of this data set to conform to Fisher's model, it will be of formal interest to carry out analyses of this type. This has been done for both the geomagnetic field directions and for the corresponding VGP's; the data are given in Table 2. Also listed are the angular standard deviation factors [*Doell and Cox*, 1965*a*] and the results of experiments to test the conformity of the distribution of these data directions to the model of Fisher [*Watson and Irving*, 1957]. At the 95%-confidence levels the distribution of geomagnetic directions is not significantly different from Fisher's model, whereas the VGP's are. The failure is in the azimuthal distribution, there being a flattening of the distribution of VGP's in a direction normal to the great circle connecting Hawaii and the mean VGP (Figure 6).

We have also carried out a two-level analysis of the data following the method given by *Watson and Irving* [1957]. The within-site precision, k_w, is equal to 304 and that between

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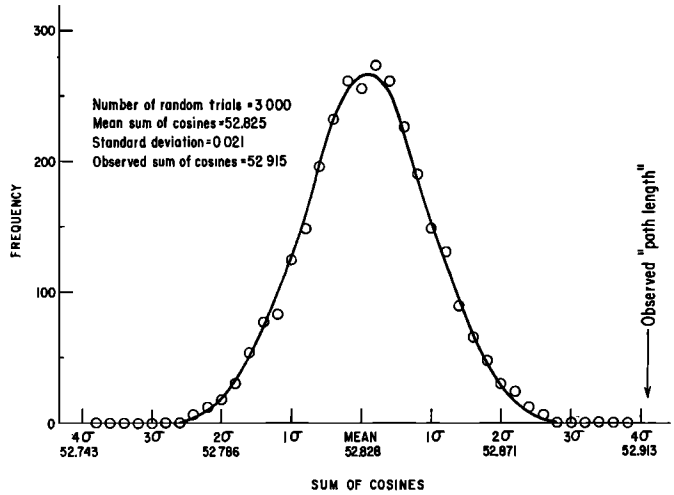


Fig. 5. Results of serial correlation test [*Watson and Beran*, 1967]. See text for details.

sites, k_B, is 610 for a total of 419 sample cores from 54 lavas (sites). The mean angular standard deviation factor corresponding to $k_W = 304$ is $S_W = 4.65°$ where $S_W = (2/k_W)^{1/2}$. This compares favorably with the average value, 4.44°, of the 54 individual values of S listed in Table 1. In a similar manner, an angular standard deviation factor, S_B, for the dispersion of the ancient field directions about their mean can be calculated from k_B. This turns out to be 3.28°. The value of S_T cited for the analysis of directions in Table 2 (3.67°) treats each lava mean direction as a datum and hence includes errors at the within-lava level, that is, $S_T = (S_B^2 + S_W^2/\bar{N})^{1/2}$ where $\bar{N}$ is the average number of sample cores from each lava [*Doell and Cox*, 1965*a*]. With average values of $S_W = 4.44°$ and $\bar{N} = 7.76$, the angular standard deviation factor for the ancient field directions is $S_B = 3.31°$, in very good agreement with the value of 3.28° found

TABLE 2. Statistical Analyses of Flow-Average Data

Data analyzed	P_a	P_r	$\frac{K_T}{k_T}$	R	α_{95}, deg	$\frac{S_T}{s_T}$, deg	Lat. / Incl., deg	Long. / Decl., deg
Virtual geomagnetic poles	99.5	75	617	53.9141	0.78	3.23	81.5N	61.3E
NRM directions	75	75	479	53.8893	0.89	3.67	+24.1	354.8
Number of lavas $N = 54$								

Notes.

P_a, probability that azimuthal distribution of data points is non-Fisherian.

P_r, probability that radial distribution of data points is non-Fisherian [*Watson and Irving*, 1957].

K_T, k_T, between-flow precision parameter [*Fisher*, 1953], virtual geomagnetic poles and directions, respectively.

R, vector sum of $N = 54$ unit vectors.

α_{95}, 95%-confidence angle about R.

S_T, s_T, angular standard deviation of mean flow data, virtual geomagnetic poles and directions, respectively.

Lat., latitude of mean virtual geomagnetic poles.

Long., longitude of mean virtual geomagnetic poles.

Incl., inclination of mean of flow directions.

Decl., declination of mean of flow directions.

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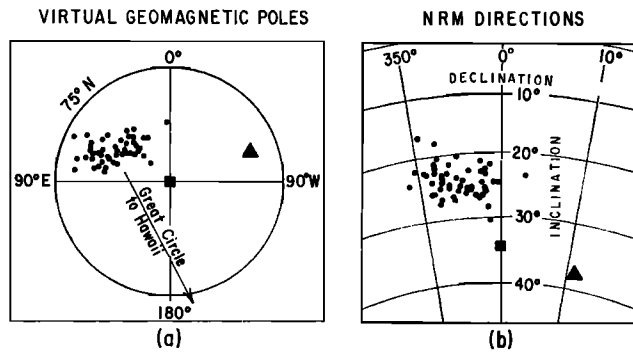


Fig. 6. Flow-average directions of natural remanent magnetization (NRM) (*b*) and associated virtual geomagnetic pole (VGP) positions (*a*) for lava flows within the Kau volcanic series. Triangle represents present geomagnetic pole position in (*a*) and present field direction in (*b*). Square represents corresponding data for axial dipole field. Portions of Lambert equal-area projections.

from Watson and Irving's methods. We may also take the within-flow dispersion into account in finding S_B, the angular standard deviation factor for scatter about the mean VGP's; this turns out to be 3.06° (=0.0534 radians), compared with the total deviation factor S_T = 3.23° given in Table 2. It is also of some interest to know the precision with which such an angular deviation factor has been found. *Cox* [1969] has given such a method, which gives $S_B = 0.0534(^{+0.0084}_{-0.0068})$ radian for the present data set.

Watson and Irving [1957] also indicate how to calculate a value of α_{95} making use of all the data—not just the flow averages. The calculated value is 0.89°, which is identical with that cited in Table 2 for the mean of the flow means. This simply means that, in the case of the Kau lavas, each flow mean is quite accurately defined. It does not mean, however, that this value of α_{95} is a realistic measure of the 95% confidence limit for this data set, because of the strong serial correlation.

Moreover, as noted earlier, one must consider the possibility of local anomalies and how these might cause the local field to deviate from the regional field. Based on the same data as previously used [*Doell and Cox*, 1963] to estimate the effect of such anomalies, the lower limit for the 95%-confidence limit about the average regional field direction γ_{95} that can be obtained from these data is about 2¼°. The actual value is probably some degrees larger, because of the serial correlation discussed above. That is, there are far fewer than 54 statistically independent data. It seems highly likely, however, that the mean direction (or VGP) of this data set is significantly different from the present field in Hawaii or that field due to an axially symmetric dipole.

Age of the Kau Volcanic Series

In order to interpret the directions of NRM within the lavas of the Kau volcanic series in terms of geomagnetic field changes, or secular variation, it is necessary to know something about the age relations of these lavas. Unfortunately, this is very difficult, but perhaps some plausible limits can be assigned. Previously [*Doell and Cox*, 1965*a*], we discussed evidence suggesting that other volcanic sequences studied on Hawaii represented thousands to tens of thousands of years time. These estimates were based primarily upon the number of lavas that had been deposited upon the Pahala ash, dated by the C^{14} method to be no younger than 10,000 years [*Rubin and Berthold*, 1961].

The Pahala ash is not exposed in the walls of Mokuaweoweo; thus, we may take 10,000 years as the maximum age for the base of the section studied there unless, of course, the ash never existed on the top of Mauna Loa. From the present distribution of the Pahala ash [*Stearns and MacDonald*, 1946, p. 19], this possibility is unlikely. The minimum age for the top lava in the section is the date of collapse of the northwest wall of the caldera, which could have been as recent as a few hundred years ago.

Of at least equal interest with regard to the

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age of eruption of these lavas is the total time spanned from the emplacement of the lowest layer sampled to the emplacement of the present summit lava. As mentioned earlier, the lavas sampled rarely extend more than several hundreds of meters laterally. We have sketched on Figure 1 several individual lavas that can presently be recognized in the summit region, and it is seen that these also have lateral extents normal to the slopes that are typically no more than a few hundred meters. Some of the lavas in the section may be sills, but, in general, the nature of the exposures throughout the section studied is essentially the same as for those lavas presently at the surface. Moreover, if very many of the lavas were sills, then the marked serial correlation of directions noted above would be difficult to explain. Thus, we conclude that the bulk of these lavas are superposed in time as well as position. Moreover, it seems reasonable to conclude that all of the lava flows accumulated in a manner similar to those that are presently lying on the surface.

Since 1832 there have been some 35 eruptions from Mauna Loa—17 from the summit region and the remainder from the flanks [*Stearns and MacDonald,* 1946, p. 81 (plus the 1949 and 1950 eruptions)]. This works out to be about one eruption each 4 years for the entire volcano during the last 146 years, or one each 8½ years for the summit region. We also note that the circumference of the present caldera is about 11 km. Now if 50 superposed lavas, as studied here, is a typical number for a 150-meter-thick section at any place on the caldera rim, and if each lava had a lateral extent of some 500 meters, then it would take of the order of 1100 lava flows to build the upper 150 meters of Mauna Loa. At the present rate of summit activity, the time needed to build this thickness would be about 9000 years. It has been suggested [*Wentworth et al.,* 1961] that the rate of eruption of Mauna Loa since 1780 has been anomalously high. If this is so, the 9000-year estimate must be taken as a minimum value.

Clearly, other more direct methods of measuring the ages of these lavas are desirable, but none are available. At present, the best guess is that the paleomagnetic directions of the Kau volcanic series lavas represent some thousands of years of geomagnetic field behavior and that this 'record' is younger than 10,000 years old.

Secular Variation in the Central Pacific Ocean

There are several features of the present set of data that are of interest in discussing the behavior of the ancient geomagnetic field in Hawaii. First, the overall angular standard deviation factor for VGP's is very small, 0.053 radians, compared with a mean value of 0.21 radians for the 112 Hawaiian lavas previously studied [*Doell and Cox,* 1965*a*]. These latter data, however, undoubtedly represent a much longer overall period of time than the Kau lavas. Second, the azimuthal distribution of VGP's is not uniform, whereas the corresponding directions are. This is compatible with a model in which the field direction of a fixed geocentric dipole is perturbed by time variations of more local sources (presumably, the changing nondipole field). In the case of the other 112 Hawaiian lavas, the opposite is true (VGP's are more evenly distributed azimuthally than field directions), although the effect is not as marked in all the other volcanic series studied [*Doell and Cox,* 1965*a*, Table 7]. A third feature to note is the strong serial correlation of directions. This can only indicate that successive paleomagnetic directions are following systematic changes in the geomagnetic field. If the changes were due simply to errors in the paleomagnetic determinations, one would expect no serial correlation at all.

There are three possibilities to consider as causes of the changes observed: (1) nondipole field changes, (2) dipole field changes, and (3) local magnetic anomaly changes. We have already shown that (2) is unlikely because of the nonazimuthal symmetry in VGP positions. *Decker* [1963] has shown that the local field can be divergent up to 12° from the regional field direction very near the rims of lava craters (Kilauea Iki). However, these anomalies fall off very quickly with distance away from the rim. One might suppose a history of caldera formation and filling, immediately adjacent to the section studied, to account for all the changes observed. But this seems highly unlikely, and indeed, the physical nature of the lavas in the section studied indicates that they were formed, not at the base of a caldera wall, but rather some distance downslope from a source or vent. We conclude, therefore, that the changes in

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geomagnetic field directions measured in the Kau volcanic series are due to changes in the nondipole portion of the field.

Actually, it is not the fact that there *are* sequential changes in these data that is unusual, but rather that the changes are so small. This is especially so if, as we have argued above, several thousand years of history are represented in these data.

Thus it appears that the lavas were erupted at a rate greater than that characteristic of changes in the nondipole field, that the magnitude of the nondipole field in this region is anomalously low, and that the total time of eruption of the Kau volcanic series was much less than the characteristic 'period' for dipole wobble.

Various models for geomagnetic field variation [*Cox*, 1962; *Creer*, 1962; *Irving*, 1964; *Cox and Doell*, 1964] have been developed to compare with paleomagnetic data, and in all of these works the present relative lack of nondipole field components in the central Pacific has been discussed. In previous reports [*Doell and Cox*, 1963, 1965*a*], we have presented data and tentative arguments suggesting that this 'window' in the nondipole field is a relatively permanent feature, having existed throughout most of the last, or Brunhes, polarity epoch.

In a summary of paleomagnetic studies of Brunhes age lavas from Hawaii, the Galapagos Islands, Alaska, and the western United States [*Cox and Doell*, 1964], it was shown that the maximum 'wobble' of the main dipole during this time could be described by an angular standard deviation factor $S_D = 0.2$ radians. This is so because total angular deviation factors S_T for Hawaiian and Galapagos Island VGP's have values close to 0.2 radians, and, hence, if any of this dispersion is due to nondipole field changes in these two regions, then S_D would have to be smaller.

The results from the Kau lavas may help refine this estimate. 'Periods' associated with secular variation due to westward drift of the nondipole field and waxing and waning of individual features are in the range of 10^2 to 10^3 years [*Cox and Doell*, 1964]. Thus, the Kau lavas studied should represent at least several such 'periods,' and a careful study of the changes in direction from lava to lava (Figure 2) suggests that this is indeed so (giving some confirmation that several thousands of years for eruption of these lavas is realistic). Therefore, the total dispersion for the Kau data, $S_T = 0.053$ radians, may be taken as a reasonable estimate of dispersion due to changes in the nondipole field during this time interval. The value $S_N = 0.053$ radians is, of course, a maximum estimate, since any other source of variation will also contribute to the total variation, making the contribution due to the changing nondipole field less.

If we further assume that this value for S_N is typical of the entire Brunhes epoch, then, using the value of $S_T = 0.21$ radians (12.0°) [*Doell and Cox*, 1965*a*, Table 7*b*] for the 112 Hawaiian lavas previously studied, we find $S_D = (S_T^2 - S_N^2)^{1/2} = 0.20$ radians as the deviation factor due to the wobble of the main dipole. From *Cox*'s [1969] estimates, the range of S_D (based on $N = 112$) is between 0.224 and 0.185 radians.

In an analysis of archeomagnetic data, *Kawai and Hirooka* [1967] and *Kawai et al.* [1965, 1967] have presented arguments that the geomagnetic dipole axis has undergone a 'wobbling' motion during the last 2000 years. Furthermore, they suggest that this wobble is due to the superposition of two circular movements: a 1500-year cycle, 1000-km-radius motion about the rotation axis (i.e., movement along the 80° parallel) and a 500-year cycle, 400-km-radius motion whose center lies on the locus of the larger motion. The former rotation is anticlockwise, the latter clockwise. If a set of paleomagnetic data randomly sampled, or even partly sequentially sampled, such a motion for an interval of 1000 years or so, we should expect a distribution of VGP's centered near the rotation axis with an angular dispersion factor of the order of 0.17 radians (10°). The data from the Kau volcanic series are clearly in conflict with this picture. There appear to be three possible explanations for this conflict: (1) The period of time covered by the Kau lavas does not overlap the last 2000 years; (2) the time spanned by the Kau volcanic series is much shorter than the 9000 years suggested above; or (3) the movements analyzed by *Kawai and Hirooka* [1967] and *Kawai et al.* [1965, 1967] are due to the nondipole portion of the field rather than the dipole portion, and the nondipole portion is missing in the region

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of the central Pacific. The first explanation may, in fact, be quite reasonable, since the mean of the Kau VGP's is far from the present geomagnetic pole (Figure 6). However, even if this were so, it would appear that motions of the dipole during eruption of the Kau lavas were vastly more subdued than during the last 2000 years. If the estimate of 9000 years is too long by as much as an order of magnitude, one would still expect to see, on the basis of the Kawai model, considerably more motion than is present in the Kau data. The last explanation can only be tested by obtaining more archeomagnetic data (i.e., well-dated results for the last few thousands of years) from different localities, especially from the central Pacific.

The conclusions arrived at above must certainly be considered as tentative. They depend rather strongly on the assumption that the Kau lavas in the wall of Mokuaweoweo represent a period of time long enough to have averaged secular variation due to changes in the nondipole field but not long enough to have sampled a change in the orientation of the main dipole. These assumptions presently seem the most reasonable, but, without absolute age control, they cannot be verified. The only other reasonable explanations for the small variation in paleomagnetic directions of the Kau volcanic series are that the period of time represented by these data is not typical of, say, the entire Brunhes epoch or that only a very short period of time is represented. As argued above, the latter explanation is not very likely; only more data can decide whether the former is a probable explanation. Meanwhile, data presently available are most compatible with a very low nondipole field variation in the central Pacific for an extended period of time, just as it has been during the period of direct observations.

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